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# Preface

In order to keep the machine in good operating condition and to prevent and minimize the occurrence of accidents, regular inspection and maintenance of the machine should be insisted upon. Anyone who operates and maintains the machine must carefully read and conscientiously carry out the maintenance and servicing provisions of this manual.

The technical maintenance of this equipment is divided into daily maintenance and parts maintenance, and the operators need to strictly abide by the operating procedures and daily maintenance system. When the machine tool failure should pay attention to protect the scene, and to the maintenance personnel to truthfully respond to the fault condition, in order to facilitate the maintenance personnel to analyze and diagnose the cause of the failure, and to eliminate.

# 1 routine maintenance

the power switch to achieve neatness, cleanliness, lubrication and safety.

## 1.1 Pre-opening check

① Check the air supply triplex for lack of oil, if necessary, follow the lubrication instruction chart and use clean lubricant for refueling.

② Check that the power switch looks and functions well and that the grounding device is complete.

③ Check whether the screws, nuts and handles of each part are loose or missing, if found, they should be tightened and made up in time.

④ Check that the electric cabinet doors and electrical safety devices of the machine are in good condition.

## 1.2 Checking during startup

① Observe the sensitivity, reliability, temperature rise, sound and vibration of motors and electrical components.

② Observe whether there is an alarm display on the touch panel.

③ Observe the transmission of the drive screw.

④ Observe the working condition of the row drill bit.

⑤ Observe the operation of each pneumatic part of the machine.

⑥ Observe whether each cooling fan on the electric control cabinet works normally.

(vii) The door of the electric control cabinet should be opened as little as possible, and it is not permitted to open the door of the electric control cabinet while the machine is in use.

## 1.3 Maintenance after shutdown

① Clean the machine and thoroughly remove any remaining wood chips from the machine.

② Wipe off dust and oil on the exterior of the machine with a rag and cleaner.

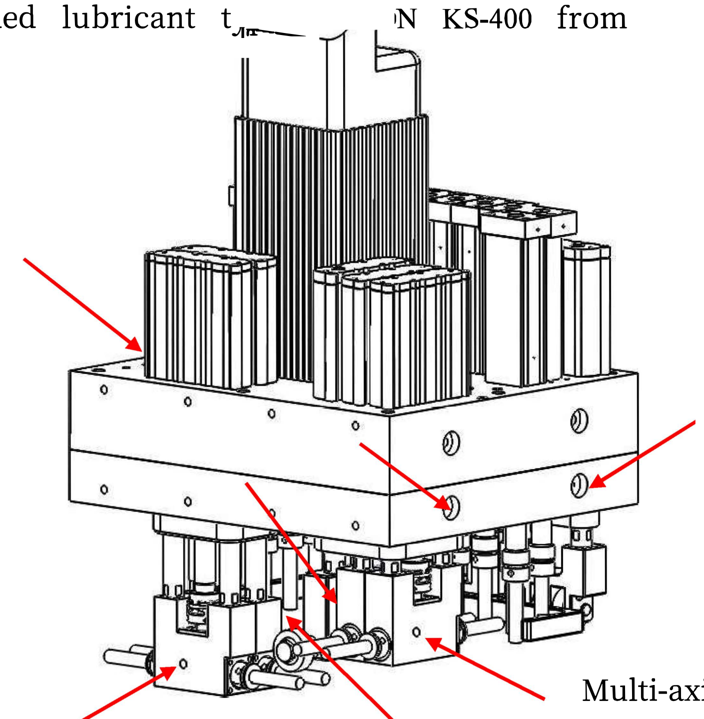
③ Sweep garbage from the ground around machine tools, organize workplace items, and keep the environment clear.

④ Place each operating handle (switch) in neutral (zero position) and pull downthe power switch to achieve neatness, cleanliness, lubrication and saf ety.

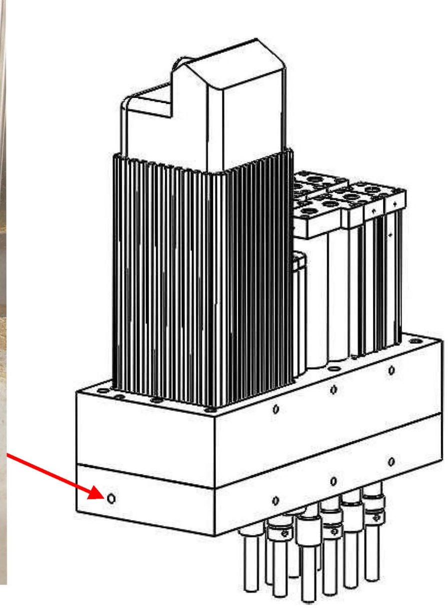
## 4 Maintenance of Multi-Axis Drilling Unit (Upper and Lower)

Lubricate the seven oil injection ports on the Multi-Spindle Drilling Unit (top) shown in the arrow every 40 hours of operation,

Multi-spindle drilling unit (lower) 1 oil filling nozzle. Oil filling volume: 5cc / time. Hint: Clean the nozzle before lubrication. Dirt in the nozzle is prohibited! Recommended lubricant t N KS-400 from Germany.



Multi-axis drilling unit (top)

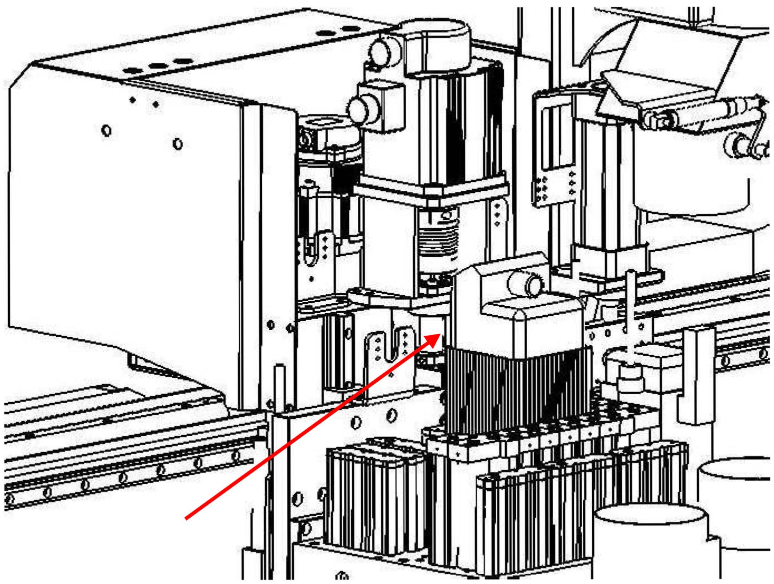



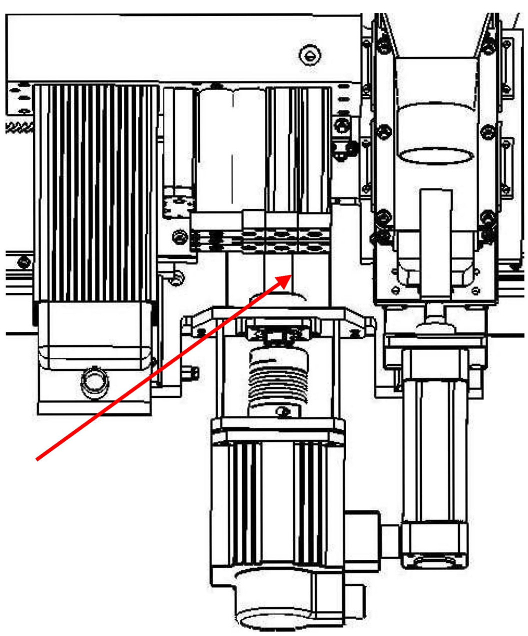
Multi-axis drilling unit (below)

## 5 Ball Screw Maintenance

Clean the dust and other debris attached to the Z-axis ball screws (upper and lower) every shift. The lubrication pump (see No.11) automatically supplies lubricating grease to continuously lubricate the ballscrews during normal operation of the machine.

If the machine is not used for a long time, such as more than one week, you need to manually apply lubricant in advance to prevent drying and rusting. Recommended oil: Great Wall Guide Oil 68#.

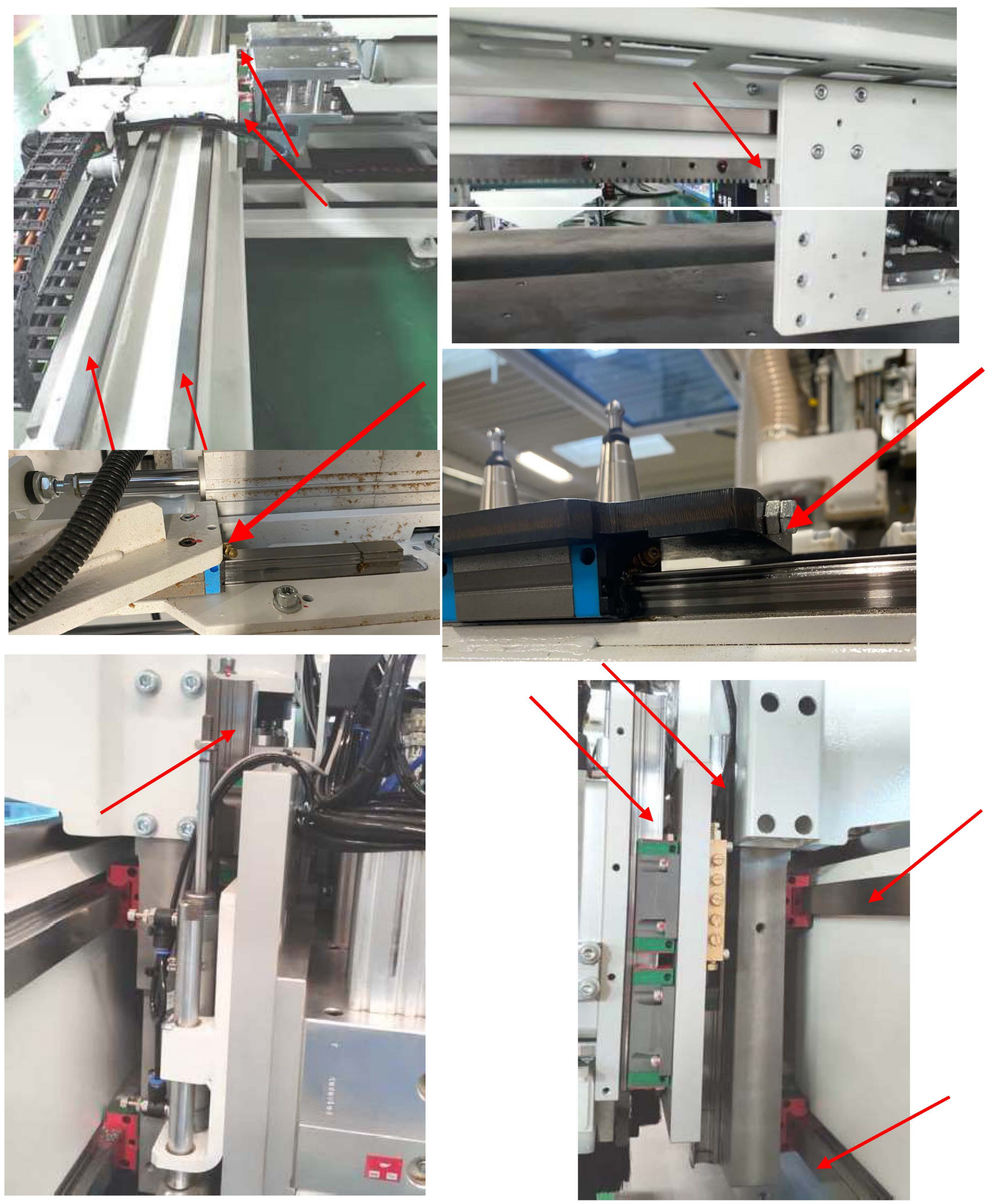




## 6 Maintenance of guideways and sliders

Dust and other debris attached to the guideways (X, Y, Z axes) are cleaned every shift. Lubricating grease is automatically supplied by the lubrication pump (see No.11) during normal operation of the machine and continuously lubricates the guide rails by means of the slider.

If the machine is not used for a long time, such as more than a week, you need to manually apply lubricant in advance to prevent drying and rusting. Recommended oil: Great Wall Guide Oil 68#.

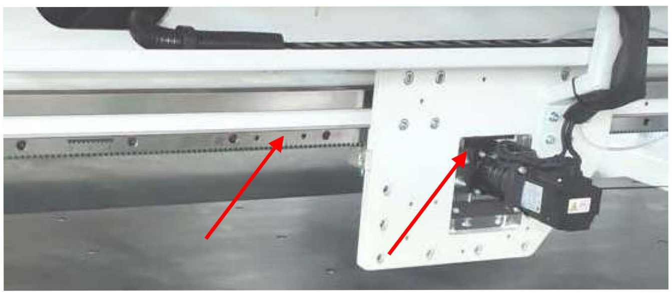


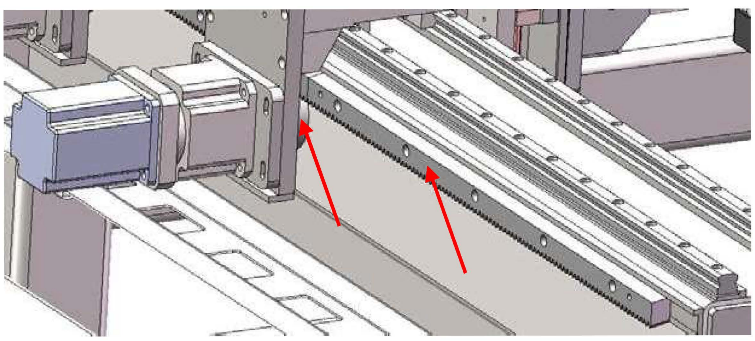
## 7 Maintenance of motor gears and racks

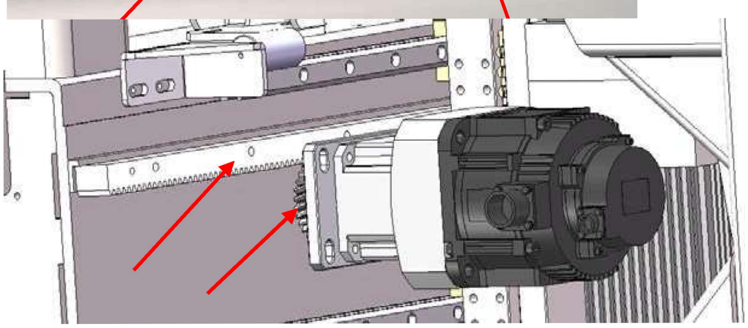
Clean dust and other debris attached to motor gears and racks (X and Y axes) every shift.

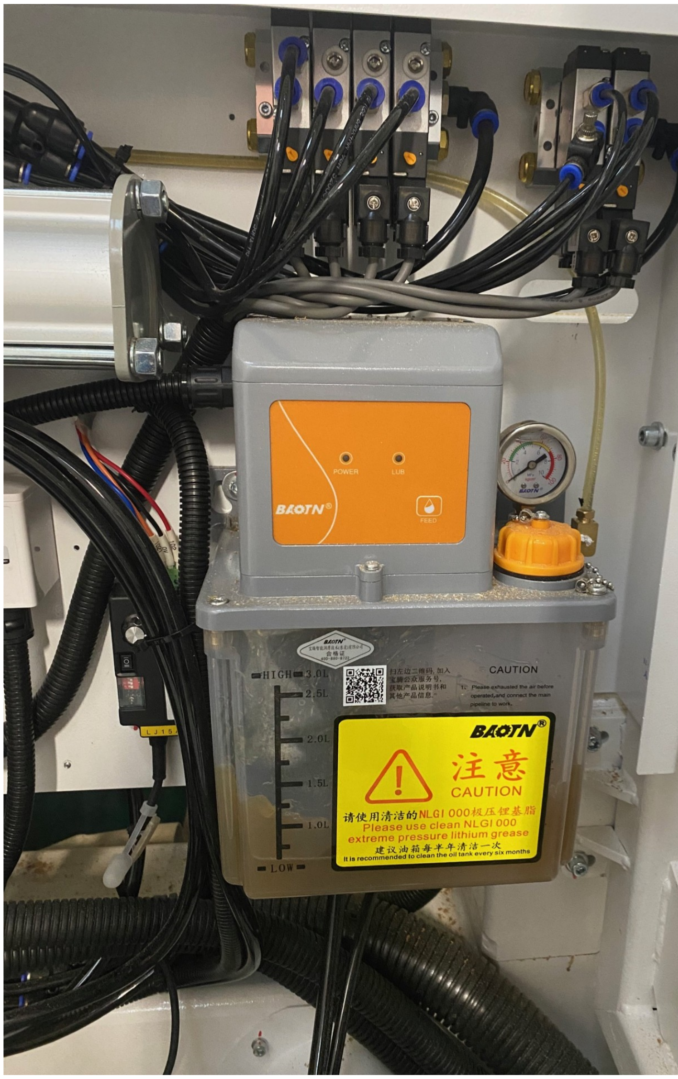
Apply grease evenly to the surface of motor gears and racks with a brush. Recommended oil: Great Wall Guide Oil 68#.

Motor gears or racks that are too dry mean inadequate lubrication and should be relubricated or lubrication intervals shortened. **Maintenance intervals: once a week**





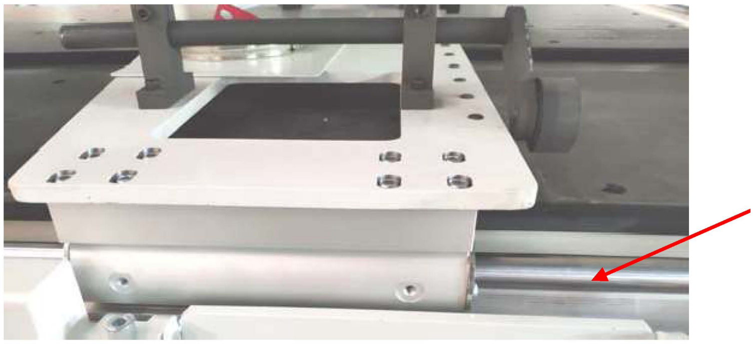



BAOTN®
POWER
LUB
FEED
HIGH
3.0L
2.5L
2.0L
1.5L
1.0L
LOW
扫左边二维码，加入
宝腾公众服务号，
获取产品说明书和
其他产品信息。
CAUTION
1、Please exhausted the air before operated,and connect the main pipeline to work.
BAOTN®
注意
CAUTION
请使用清洁的NLGI 000极压锂基脂
Please use clean NLGI 000
extreme pressure lithium grease
建议油箱每半年清洁一次
It is recommended to clean the oil tank every six months

超越
Chaoyue
Model
HP-5
Output volume
5cc/st
Discharge pressure
5Kgf/cm²
Effective capacity
650cc
操作说明
1. 必须使用20-150cSt之清洁润滑油。
2. 操作时，只能压动手柄一次，待供油完毕（手动自动复原），方可进行下次动作。
3. 新机初次使用时请先压动手柄两次，以便润滑油充满整个管路。
4. 严禁将手柄下压后上拉。

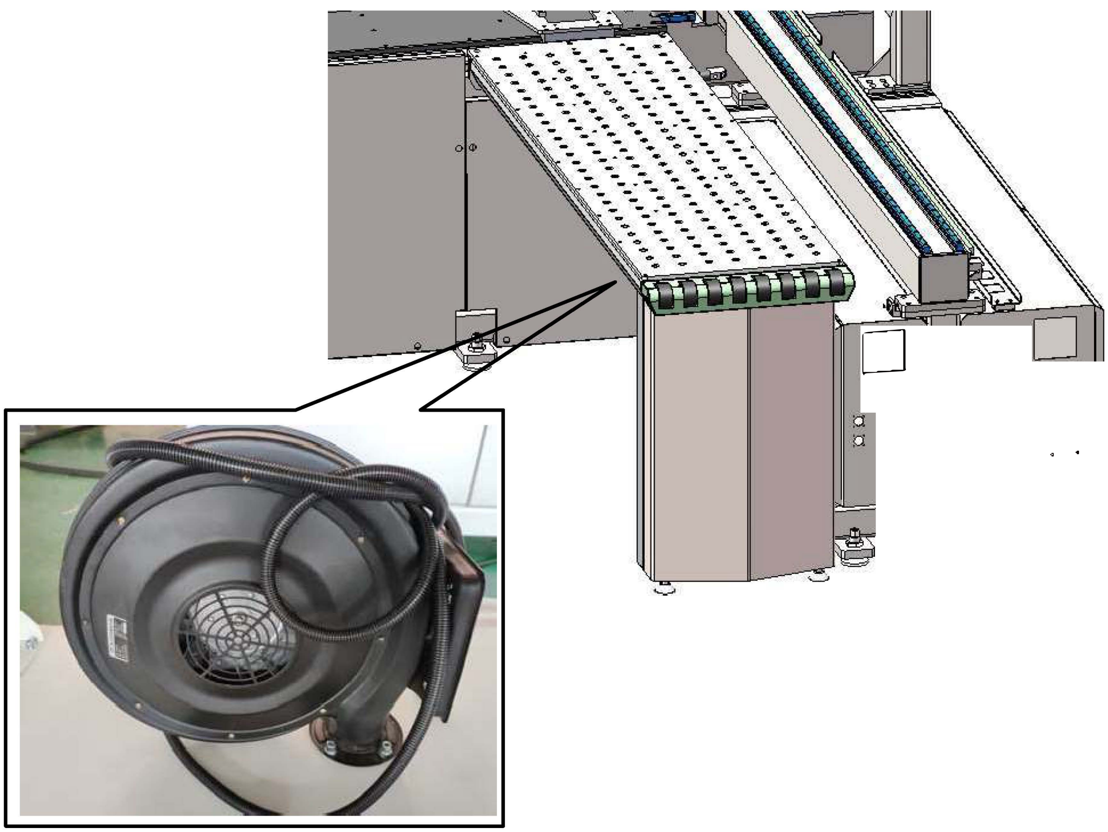
## 8 Maintenance of mobile sliders and round support rails

Clean dust and other debris adhering to the round support rails every shift.
Apply grease evenly to the surface of the round support rail with a brush.
Recommended oil: Great Wall Rail Oil 68#. **Maintenance cycle: once a week**



## 9 Maintenance of air flotation table and blowers

Clean dust and other debris attached to the air flotation table every shift.
Check the size of the airflow from the blower, if the airflow decreases, it needs to be overhauled to ensure normal use. **Maintenance cycle: once a month**



## 10 Maintenance of the air supply triplex

1. When adding special pneumatic oil to the oil mist in the air supply triplex, follow the requirements below:

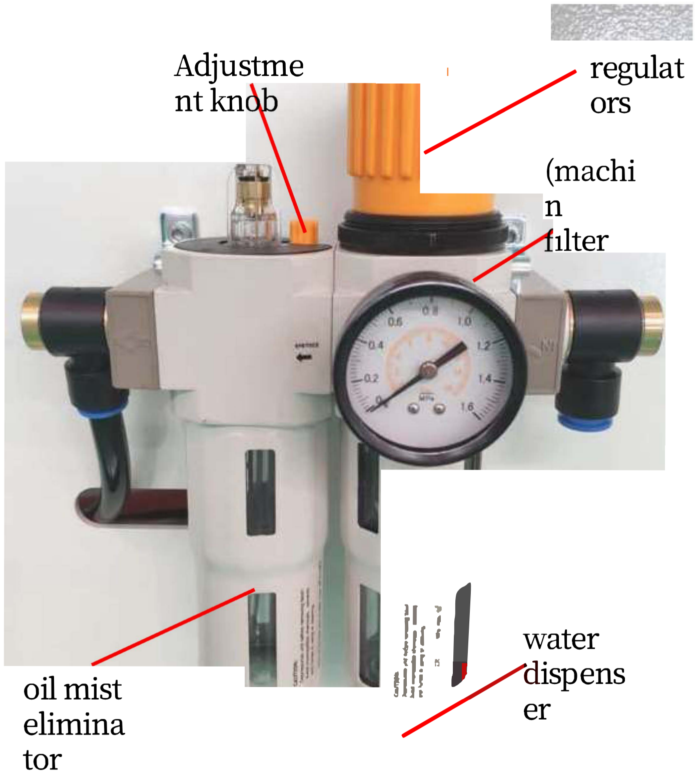
- Turn off the air supply and exhaust the working air pressure; rotate the oil mister counterclockwise;
- The volume of oil should not exceed 2/3 of the oil atomizer when refueling.

recommended oil: ISO VG-32 or the same grade. 2. The regulator is operated as follows:

- Pull up on the regulator handwheel to put it in the unlocked position;
- Slowly pressurize the entire system by shutting off the air supply as far as possible in the "one" direction;
- Turn the regulator handwheel in the "+" direction until the desired pressure is indicated on the gauge;

3. Adjust the pneumatic oil flow using the adjusting knob when appropriate. For example, when the compressed air flow rate is 1000L/min, adjust the oil cup to 1-2 drops of oil per minute.
4. When the condensate in the pressure regulating filter reaches the filter element at about 10 mm, rotate the water drain counterclockwise so that the condensate can flow out in a smooth manner;

- When the system pressure decreases significantly, the filter element needs to be cleaned or replaced in time.



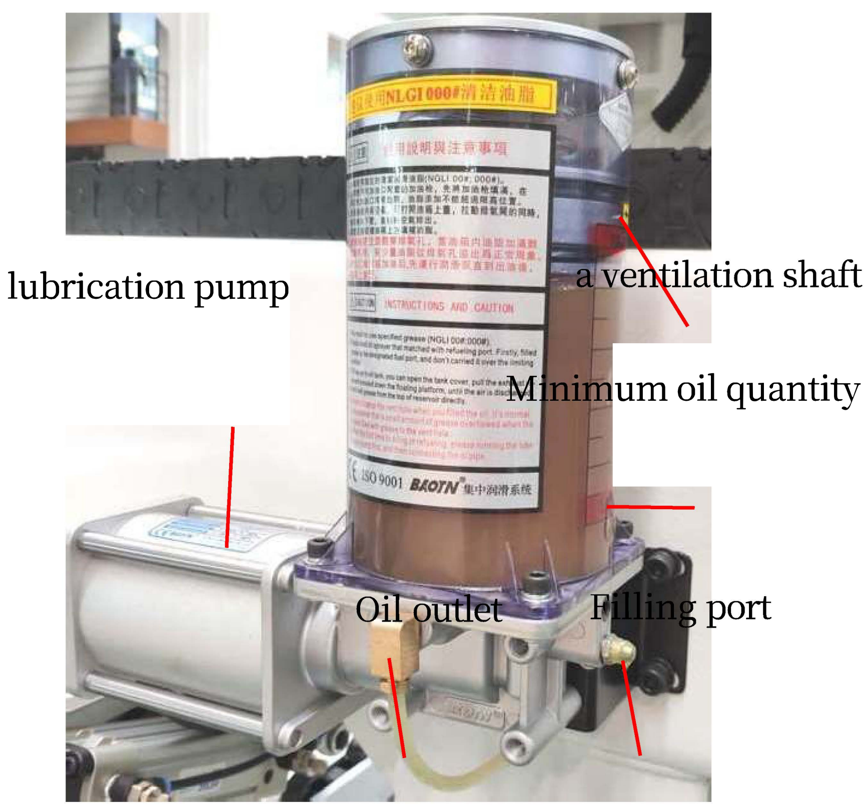

**Maintenance cycle: once a week**

## 11 Maintenance of lubrication pumps

Regularly check the oil level of the oil tank, weekly inspection, below the minimum oil level, you need to lubricate the pump in a timely manner to inject lubricating grease to ensure the normal lubrication of all parts of the machine.

Specific instructions and precautions for use are listed below:

1. Use a specified clean lubricating grease (NLGI 00#;000#), Mobilux grease EPO is recommended.
2. Fill the grease from the designated grease filler, grease should not be added beyond the height limit position.
3. If there is air in the tank, open the top cover of the tank and while pulling the vent valve, press the float down until the air is discharged.
4. It is strictly forbidden to fill grease directly from above the storage tank.
5. When refueling, please pay attention to the exhaust hole, when the tank is full of grease to the exhaust hole, a small amount of grease overflow from the exhaust hole is a normal phenomenon.
6. Before initial oiling or after refueling, run the lubrication pump until oil comes out, then connect the fuel line. **Maintenance cycle: once a week**



## 12 electronic control cabinet

a. Cabinet inspection and dust removal

➢ Maintenance content:

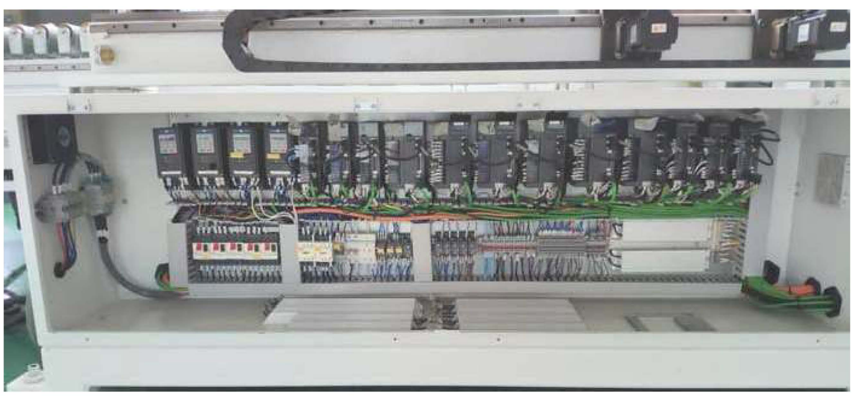
① Check that the doors, covers, locks and gaskets around the door frames of electrical cabinets are in good condition.

② Remove the accumulated dust in the electrical cabinet and keep the surface of the electrical components clean.

③ Check the components for looseness, oxidation and failure, which need to be promptly notified to the professionals for repair or replacement.

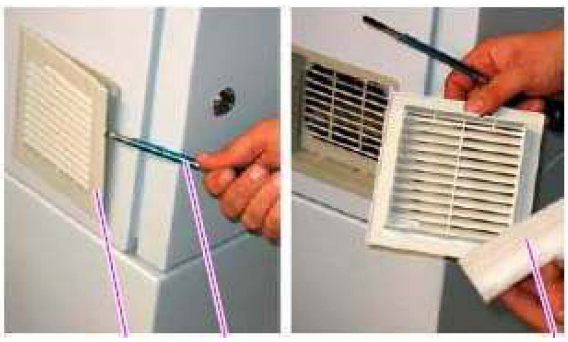
✓ CAUTION: Do not use high-pressure air to blow wash the switchgear to prevent electrical parts from loosening and foreign matter from entering the inside of the electronic components.

➢ Maintenance intervals: Monthly inspections.



b. Cooling fan cleaning

- Maintenance content: check whether each cooling fan on the electrical cabinet works normally. Regularly check whether the air duct filter is clogged, if too much dust accumulates on the filter mesh needs to be cleaned in time. If you do not clean up in time, it will cause the temperature inside the electrical cabinet to be too high, resulting in system overheating.
- ✓ Cleaning method: Use a screwdriver to remove the fan cover, remove and clean the filter, and then install it in order.
- Maintenance cycle: regular monthly cleaning.



c. Major Component Inspection

- Maintenance content: check the servo drive, inverter, switching power supply, transformer and other major components whether there is oxidation or burnt, etc.; check the joints, junction boxes with or without oxidation, burnt, such as oxidation and burnt need to be replaced in a timely manner the corresponding components.
- ✓ Note: This operation must be maintained and inspected by an electrical professional.
- Maintenance cycle: Half-yearly inspection.

# Lubrication areas and intervals

Regular lubrication to the lubrication part is to ensure the normal operation of the machine is the key. The lubrication parts and period, see the following table.

| serial number | Lubrication | lubrication cycle | Recommended Oils |
|---|---|---|---|
| 1 | Multi-axis drilling unit | 40 hours | Germany BOTON KS-400 |
| 2 | Guides and slides | Once a week (when out of service) | Great Wall Guide Oil 68# |
| 3 | ball screw | Once a week (when out of service) | Great Wall Guide Oil 68# |
| 4 | Motor gears and racks | once a week | Great Wall Guide Oil 68# |
| 5 | Moving slider and round support rail | once a week | Great Wall Guide Oil 68# |
| 6 | Triplex Oil Mist | once a week | 1# Turbine oil (ISO VG32) |
| 7 | Turning and sliding part | once a day | Great Wall Grease 3# |
| 8 | Rust-prone areas | One week (or discontinued) | rust preventive oil |
| 9 | | | |
| 10 | | | |

# Maintenance schedule for CNC six-sided drilling equipment

| Maintenance area | Maintenance items | Maintenance content | maintenance cycle |
|---|---|---|---|
| **entire machine** | entire machine | Inspection and clearance | once a day |
| **Spindle section** | spindle | Check lubrication/cleaning | Weekly maintenance |
| **Multi-axis drilling unit section** | Multi-axis drilling unit | Check lubrication/cleaning | 40 hours maintenance |
| **Ball Screw Section** | ball screw | Check lubrication/cleaning | Weekly maintenance |
| **Guideway and slide section** | Guides and slides | Check lubrication/cleaning | Weekly maintenance |
| **Motor gear and rack section** | Motor gears and racks | Check lubrication/cleaning | Weekly maintenance |
| **Moving slider and round support rail section** | Moving slider and round support rail | Check lubrication/cleaning | Weekly maintenance |
| **Air flotation table and blower section** | Air flotation table and blowers | Check lubrication/cleaning | Monthly maintenance |
| **Air source triplex section** | Air supply triplex | Check lubrication/cleaning | Weekly maintenance |
| **Lubrication pump section** | lubrication pump | Check lubrication/cleaning | Weekly maintenance |
| **Electrical box section** | electrical box | Check lubrication/cleaning | Monthly maintenance |
| | | | |
| | | | |

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